# ГРАНИЦИ НА ФУНКЦИИ - ЗАДАЧИ

**1.** Намерете границите на следните функции:

$$\lim_{x\to\infty} \frac{\sqrt{9x^2+1}-\sqrt[3]{x^2-1}}{\sqrt[4]{x^4+1}-\sqrt[3]{x^4-1}} \tag{1}$$

$$\lim_{x\to\infty} \frac{\sqrt[3]{1+\frac{4}{x}}-\sqrt[4]{1+\frac{3}{x}}}{1-\sqrt[5]{1-\frac{5}{x}}} \tag{2}$$

$$\lim_{x\to\infty} \left(\sqrt{x^2+\sqrt{x^2+\sqrt{x^2}}}-\sqrt{x^2}\right) \tag{3}$$

$$\lim_{x\to 0} \frac{\sqrt[n]{a+x}-\sqrt[n]{a-x}}{x}, \quad a>0 \tag{4}$$

$$\lim_{x\to\infty} \left(\frac{x}{2x+1}\right)^{x^2} \tag{5}$$

$$\lim_{x\to\infty} \left(\frac{x}{x+1}\right)^{x} \tag{6}$$

$$\lim_{x\to\infty} x^2 \ln\cos\frac{\pi}{x} \tag{7}$$

$$\lim_{x\to 0} x^2\left(4^{1/x}-4^{1/(x+1)}\right) \tag{8}$$

$$\lim_{x\to 0} \left(\sqrt{1+x}-x\right)^{1/x} \tag{9}$$

$$\lim_{x\to 0} \left(1-\operatorname{tg}^2 x\right)^{\sin^{-2}x} \tag{10}$$

$$\lim_{x\to 0} \left(\ln(e+x)\right)^{\operatorname{cotg} x} \tag{11}$$

$$\lim_{x\to 0} \frac{\ln\left(1+3x+x^2\right)+\ln\left(1-3x+x^2\right)}{x^2} \tag{12}$$

$$\lim_{x\to \pi/2} \left(\frac{\pi}{\cos x}-2x\operatorname{tg} x\right) \tag{13}$$

$$\lim_{x\to 7} \frac{\sqrt{x+2}-\sqrt[3]{x+20}}{\sqrt[4]{x+9}-2} \tag{14}$$

$$\lim_{x\to 1} \frac{\sqrt[m]{x}-1}{\sqrt[n]{x}-1} \tag{15}$$

$$\lim_{x\to\infty} \left(\sqrt[3]{x^3+3x^2}-\sqrt{x^2-2x}\right) \tag{16}$$

$$\lim_{x\to\infty} \frac{\left(x-\sqrt{x^2-1}\right)^n+\left(x+\sqrt{x^2+1}\right)^n}{x^n} \tag{17}$$

$$\lim_{x\to 0} \frac{\left(\sqrt{x^2+1}+x\right)^n-\left(\sqrt{x^2+1}-x\right)^n}{x} \tag{18}$$

$$\lim_{x\to 0} \frac{x^2}{\sqrt{1+x\sin x}-\sqrt{\cos x}} \tag{19}$$

$$\lim_{x\to 0} \frac{\sqrt{\cos x}-\sqrt[3]{\cos x}}{\sin^2 x} \tag{20}$$

$$\lim_{x\to 0} \left(\frac{1+\operatorname{tg} x}{1+\sin x}\right)^{\sin^{-2}x} \tag{21}$$

$$\lim_{x\to\infty}\left(\sin\frac{1}{x}+\cos\frac{1}{x}\right)^{x} \tag{22}$$

$$\lim_{x\to\infty}\left(\sin\sqrt{x+1}-\sin\sqrt{x}\right) \tag{23}$$

$$\lim_{x\to 0}\frac{\ln\cos ax}{\ln\cos bx} \tag{24}$$

$$\lim_{x\to 0}\frac{1-\cos^{\alpha}x}{x^2} \tag{25}$$

$$\lim_{x\to 0}\frac{e^{x^2}-\cos^{\sqrt{2}}x}{x^2} \tag{26}$$

$$\lim_{x\to 0}\left(2e^{x/(x+1)}-1\right)^{(x^2+1)/x} \tag{27}$$

$$\lim_{x\to 0}\frac{e^{\sin 2x}-e^{\sin x}}{\operatorname{tg} x} \tag{28}$$

$$\lim_{n\to\infty} n^2\left(\sqrt[n]{a}-\sqrt[n+1]{a}\right) \tag{29}$$

$$\lim_{x\to 0}\frac{\arccos(1-x)}{\sqrt{x}} \tag{30}$$

$$\lim_{x\to -1_+}\frac{\sqrt{\pi}-\sqrt{\arccos x}}{\sqrt{x+1}} \tag{31}$$

$$\lim_{x\to-\infty}\left(x+\sqrt{\frac{x^3+2x^2}{x+1}}\right) \tag{32}$$

$$\lim_{x\to-\infty}\arcsin\left(\sqrt{x^2+x}+x\right) \tag{33}$$

$$\lim_{x\to 0_+}\frac{1-\sqrt{\cos x}}{x-x\cos\sqrt{x}} \tag{34}$$

$$\lim_{x\to 0_+}\left(\frac{1+\sin x-\cos x}{1+\sin 4x-\cos 4x}\right)^{1/x} \tag{35}$$

$$\lim_{x\to\infty}\frac{\ln\left(2+\sqrt{x}\right)}{\ln\left(6+\sqrt[6]{x}\right)} \tag{36}$$

$$\lim_{x\to 2}\frac{\operatorname{arctg}(2-x)+\sin(x-2)^2}{x^2-4} \tag{37}$$

$$\lim_{x\to 0}\frac{x^2\left(\sqrt[3]{1+3x}-1\right)+\sin^3 x}{1-\sqrt{1+x^3}} \tag{38}$$

$$\lim_{x\to 0}\frac{(\sin 2x-2\operatorname{tg} x)^2+(1-\cos 2x)^3}{\operatorname{tg}^7(6x)+\sin^6 x} \tag{39}$$